CS228: Introduction to Deep Learning

Lecture 2: perceptron

Instructor: Zhouxing Shi

Project (40%)

- A team project to strengthen hands-on experience in deep learning
- Each team should have 2~3 students
- Implementation:
  - AI coding is allowed.
  - Use of open-source code is encouraged but should be documented.
  - New contributions should be demonstrated.
- Deadlines are strict

Project (40%)

- Project types:
  - Engineering-focused: An application of deep learning
    - Design and implement a deep learning pipeline
    - Develop a simple demo
  - Research-focused: A new model / algorithm for a deep learning problem
    - Study existing methods and design a well-motivated new method
    - Comparison with existing approaches (negative results are fine)
    - Analysis of the method and results
  - Research-focused: A novel empirical or theoretical analysis of existing approaches
- Some project directions will be suggested during the class

Project (40%)

- **Sign up (due on 4/12, 5%): Form a team and sign up.**
  - Spreadsheet link on Canvas (Assignments -> Project team signup)
  - https://docs.google.com/spreadsheets/d/1dFs6IQ3oFAFhTA0VZ1fTB3uQrKxaO_9ik7s8riW-1wI/edit

Project (40%)

- Proposal (due on 4/19, 10%):
  - Choose a task for your project and submit a 1-page writeup.
  - 1) An introduction and the background of the task;
  - 2) An overview of existing methods, including deep learning approaches and/or others (if any);
  - 3) Datasets and metrics.
  - You do not have to propose any new method at this stage.

Project (40%)

- Milestone report (due on 5/6, 15%)
  - A new writeup, updated and extended from your proposal. Add:
  - 1) Initial progress: reproduction of one or two baseline methods (reuse existing code if possible), or an implementation of a straightforward or naive method;
  - 2) Proposal on next steps that propose and implement a new method, conduct a new analysis, or develop a new tool or demo.
- Final presentation (in-class on Week 10, 20%)
- Final report and code submission (due on 6/12, 50%)

Project (40%)

- Midterm project discussion (Friday, 5/8, optional):
  - Individual project discussions between each group and the instructor on Week 6
  - 20-min slots for Zoom meetings
  - In class + extended availability as needed

Project - compute resources

- Google Colab (https://colab.research.google.com/): notebooks with GPUs

Colab Pro

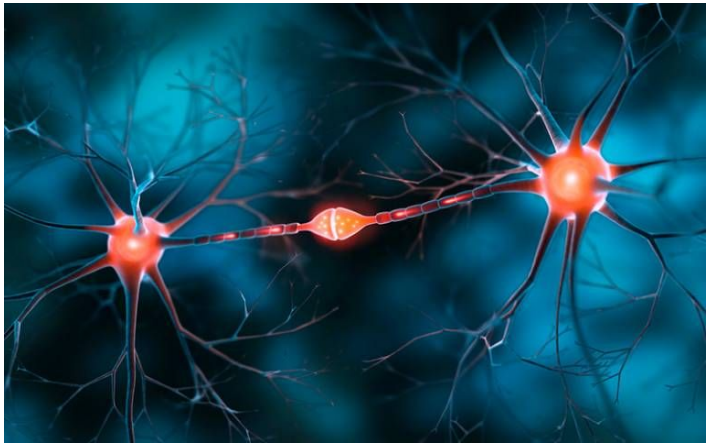
$9.99 per month

Colab Pro for Education

No cost for students and educators

- National Research Platform (NRP, https://nrp.ai): kubernetes pods with GPUs
  - Access will be set up after teams are formed

Modeling the brain



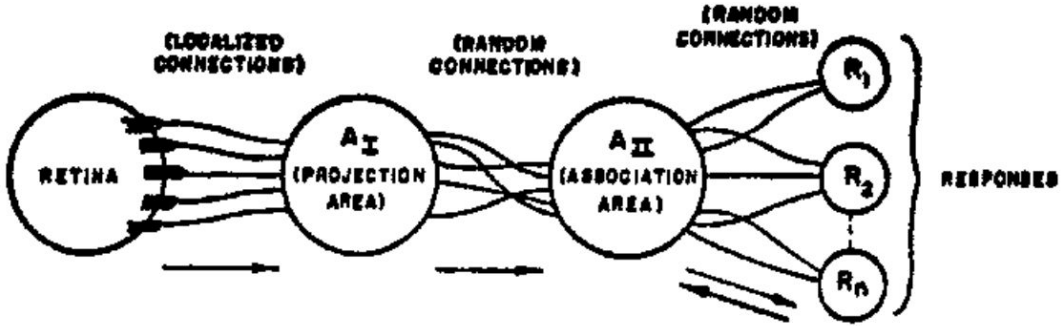
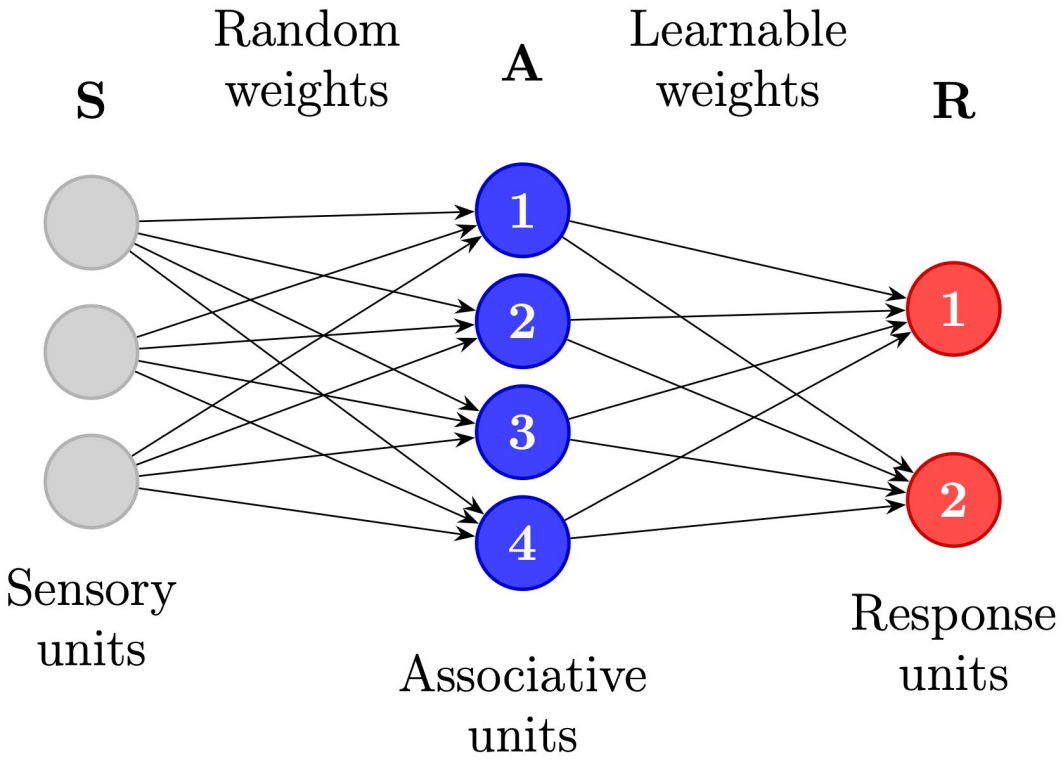


FIG. 1. Organization of a perceptron.

Rosenblatt's **perceptron** in 1958

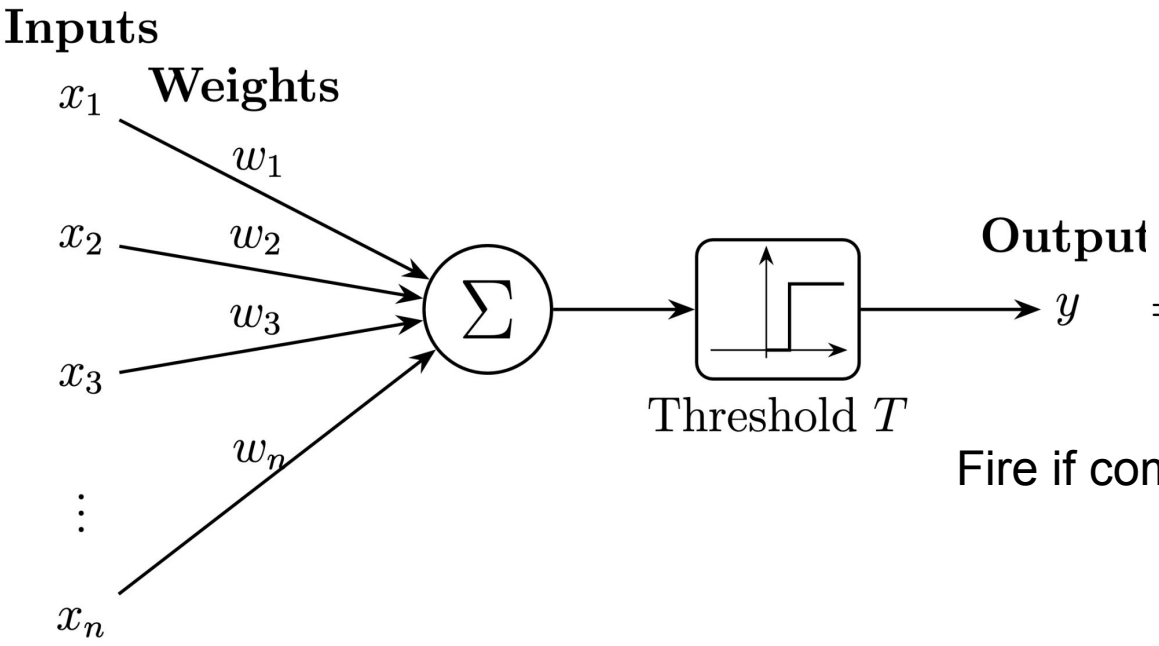
Rosenblatt's perceptron



Each of A units and R units is a **perceptron**

Perceptron

Inputs combined linearly followed by a threshold logic

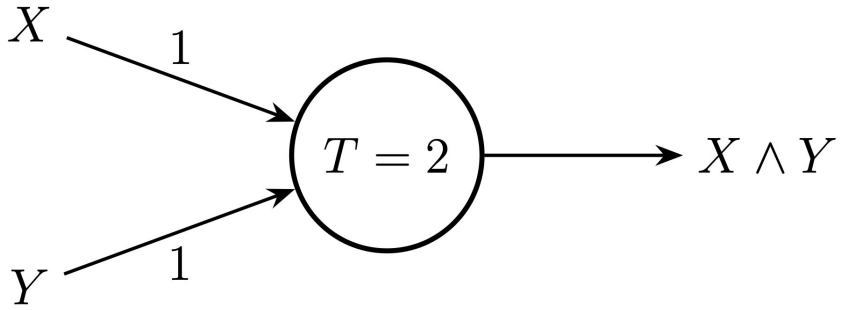


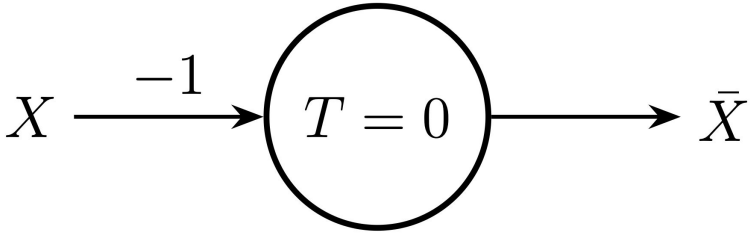
$$= \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

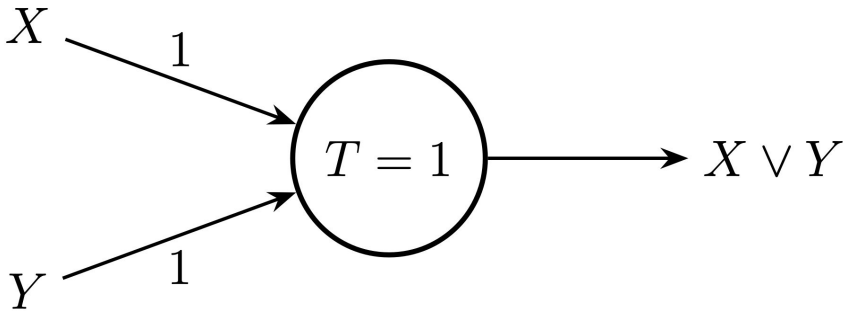
Fire if combined input exceeds threshold

Boolean operations

A perceptron can implement boolean operations

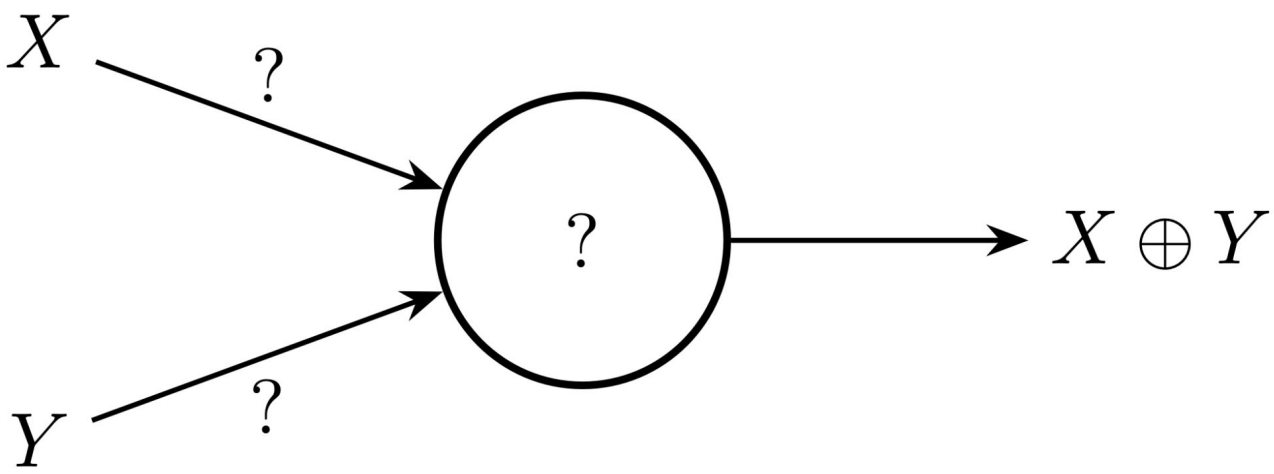






Boolean operations

How about XOR?



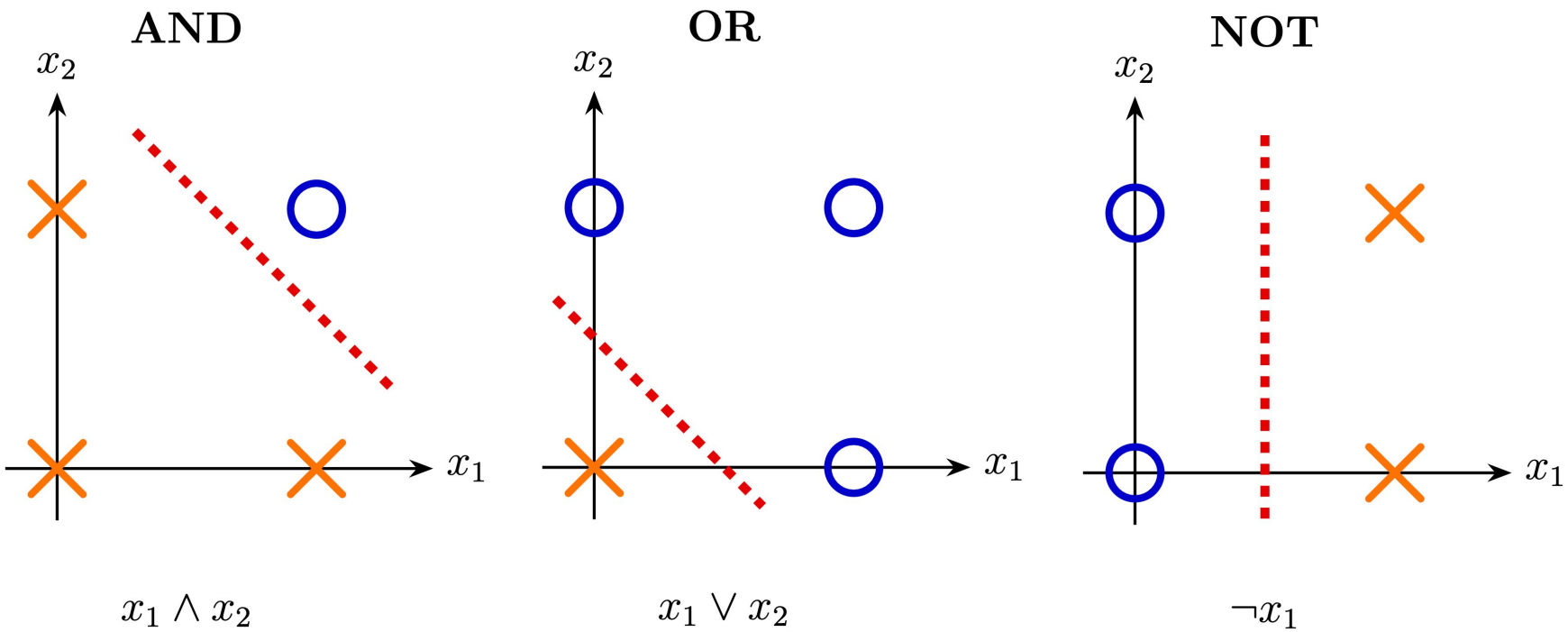
Linear separability

A single perceptron is a linear classifier

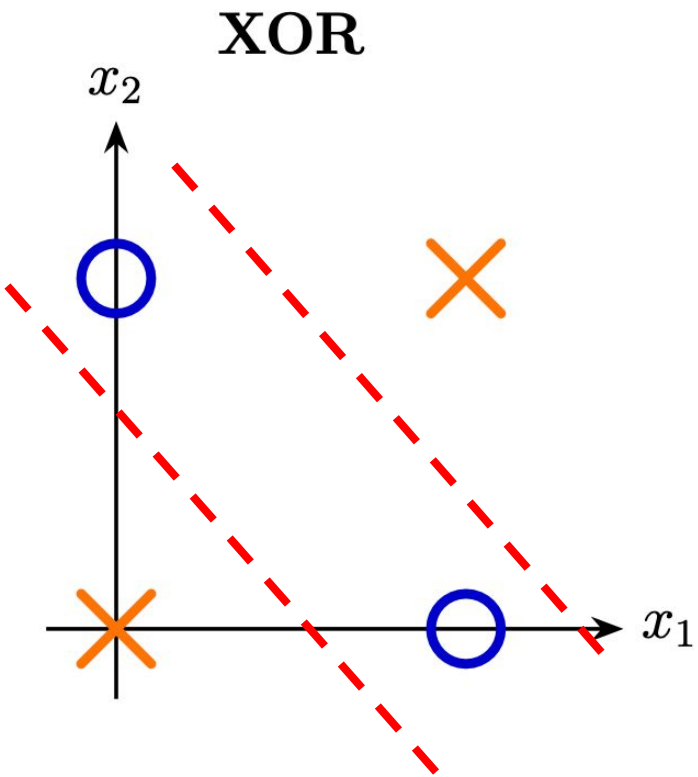
$$y = \begin{cases} 1 & \text{if } \boxed{\sum_i w_i x_i - T} \geq 0 \\ 0 & \text{else} \end{cases}$$

Affine function of inputs

Linear separability



Linear separability

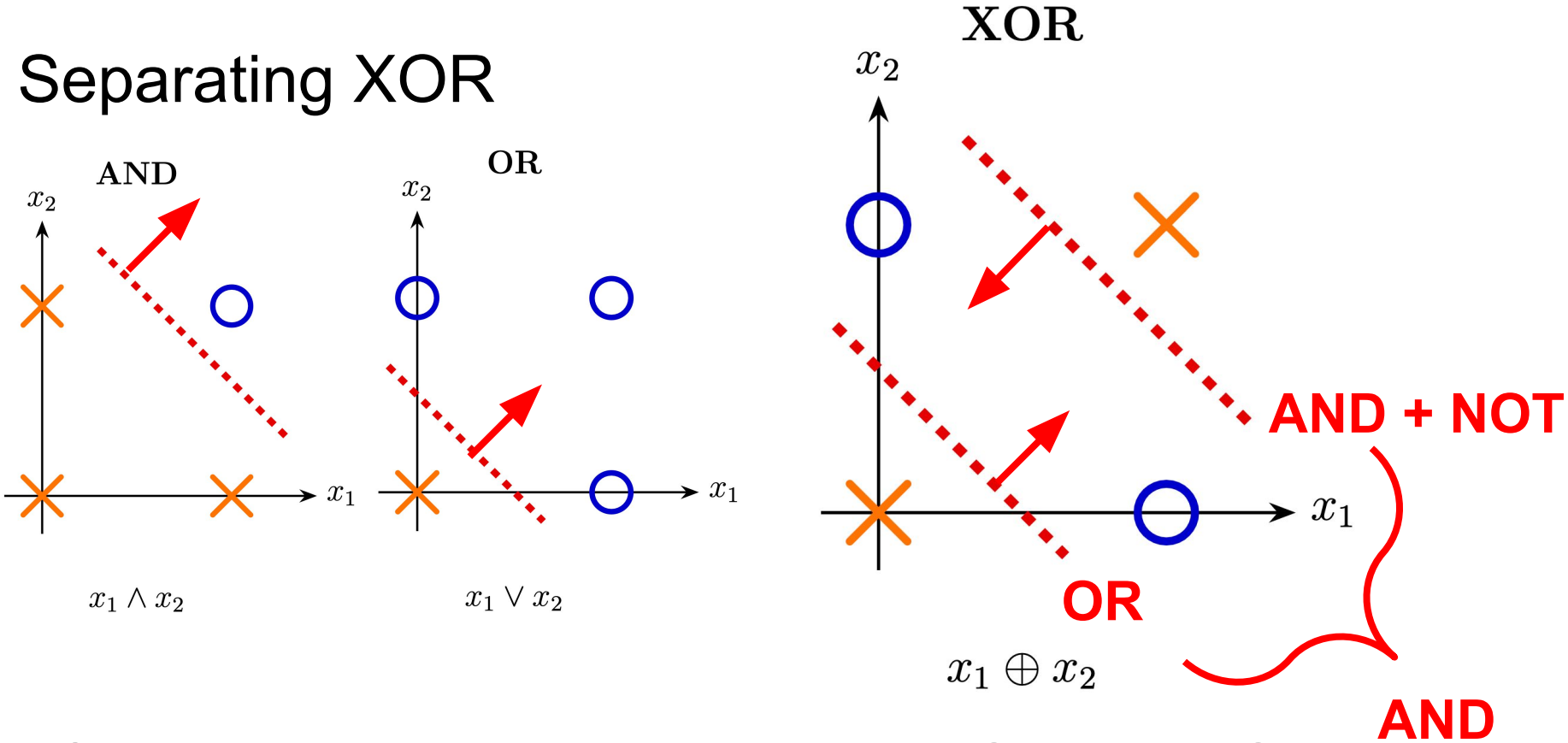


Not linearly separable!

A single perceptron cannot model nonlinear decision boundaries

Two lines can separate XOR

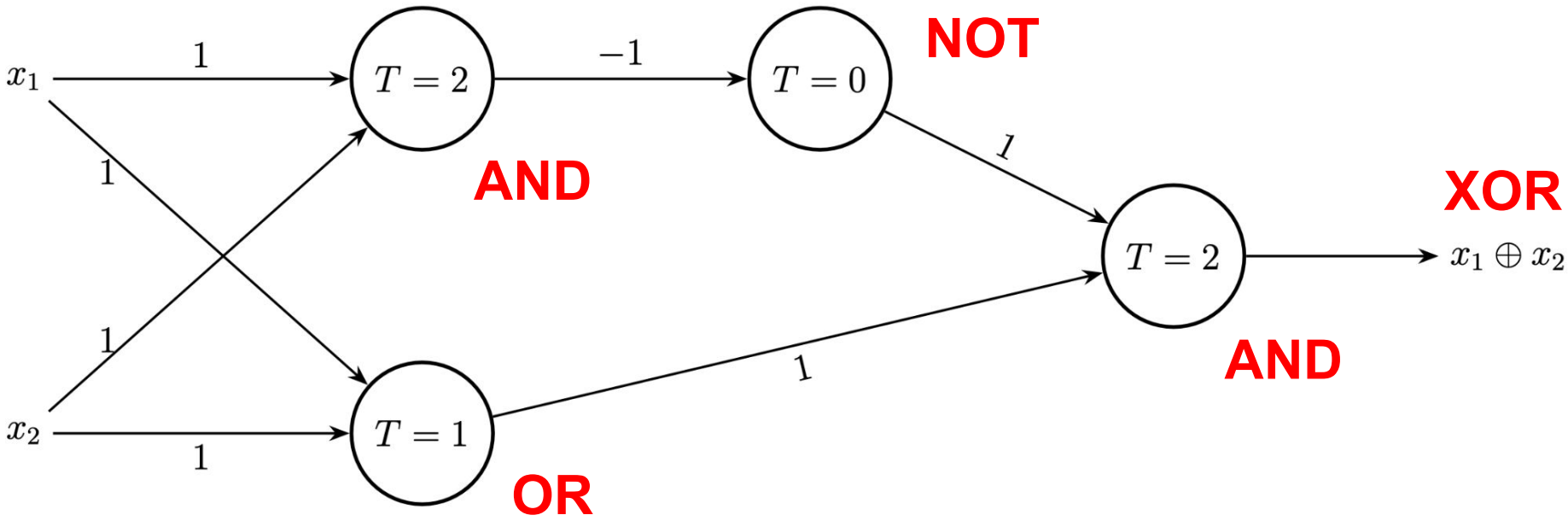
Separating XOR



XOR is separable by **combining AND, OR, and NOT**

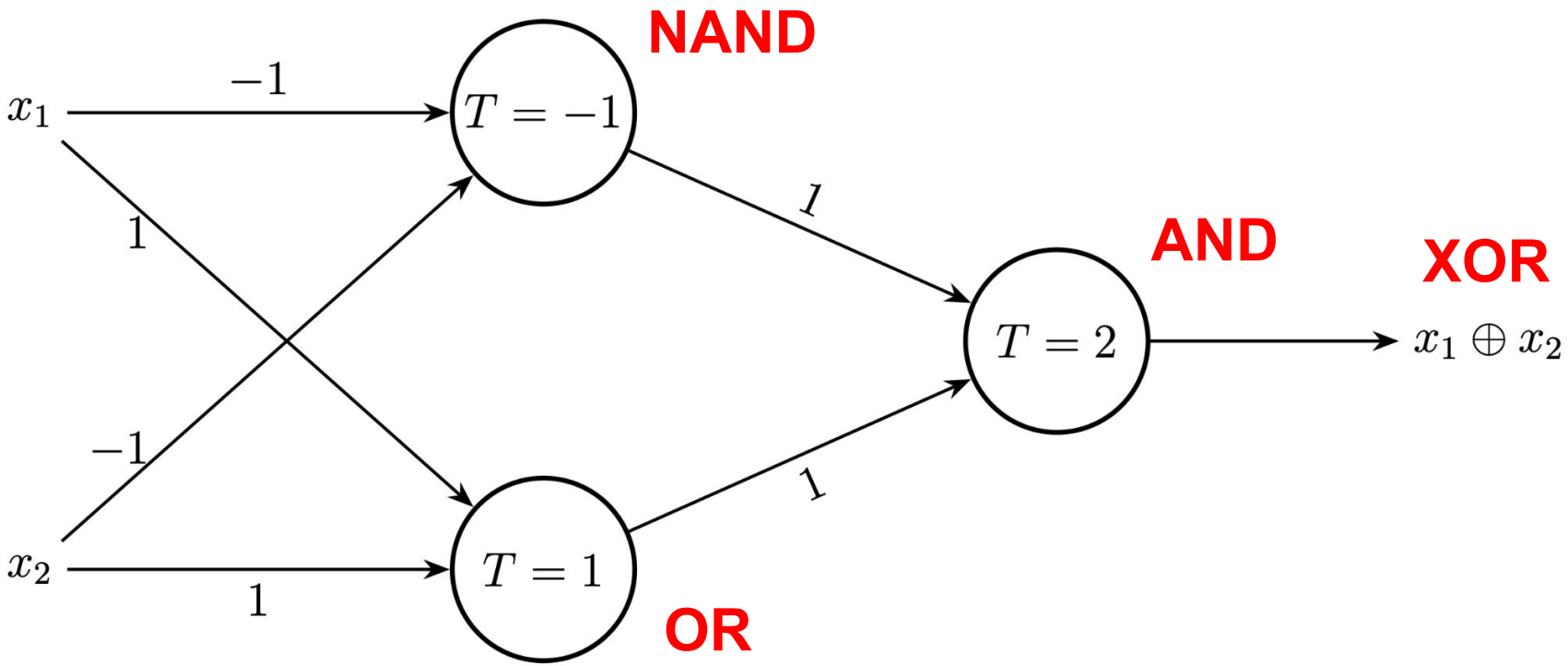
Multi-layer perceptron (MLP)

To represent any boolean functions



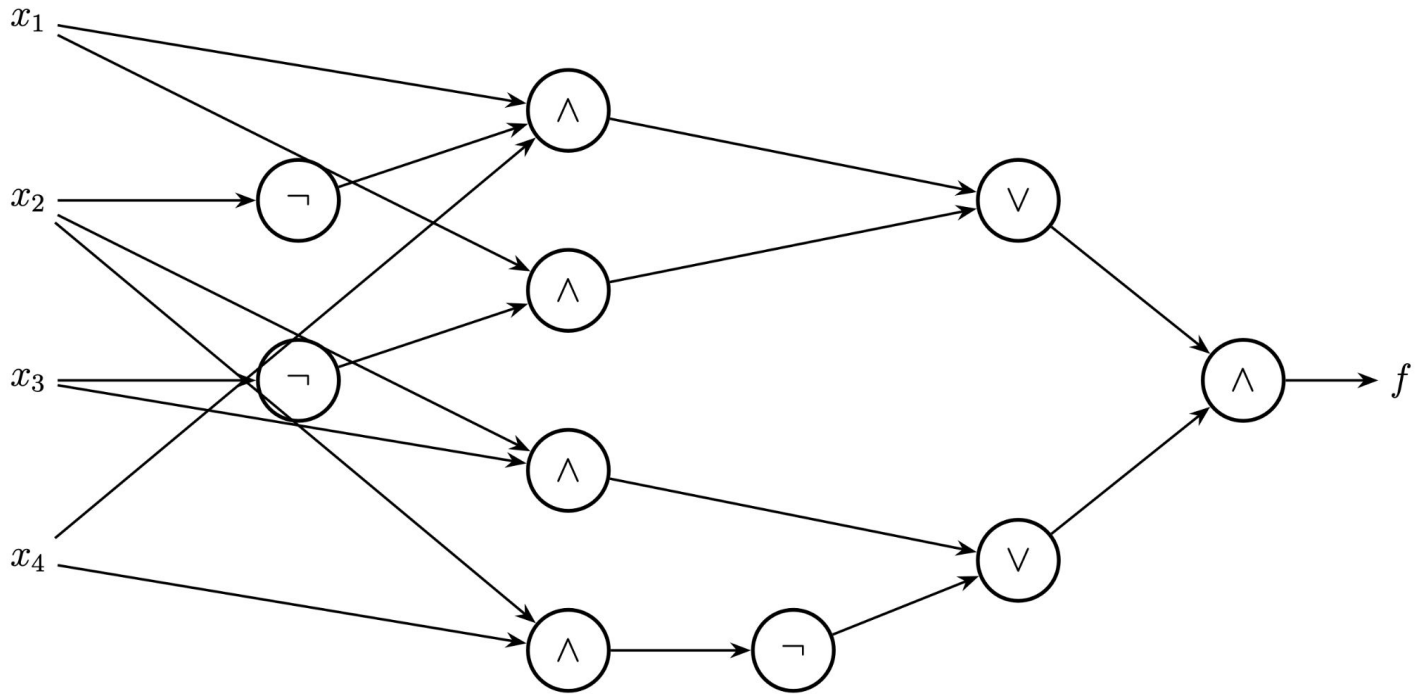
Multi-layer perceptron (MLP)

To represent any boolean functions



A more complex boolean function

$$f(x_1, x_2, x_3, x_4) = \Big((x_1 \,\&\, \bar{x}_2 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_3) \Big) \,\&\, \Big((x_2 \,\&\, x_3) \mid \overline{x_2 \,\&\, x_4} \Big)$$



A more complex boolean function

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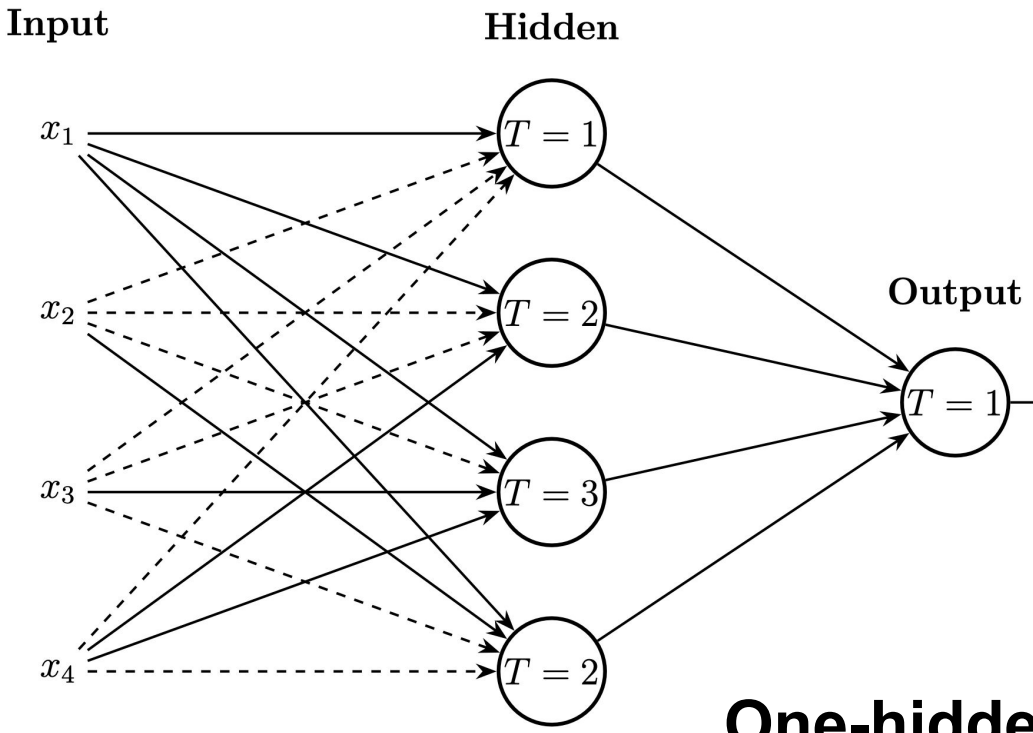
Truth table (showing rows outputting 1 only)

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$\begin{aligned} f = {} & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\ & \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\ & \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\ & \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \end{aligned}$$

Any boolean function can be represented by a **truth table**

A more complex boolean function



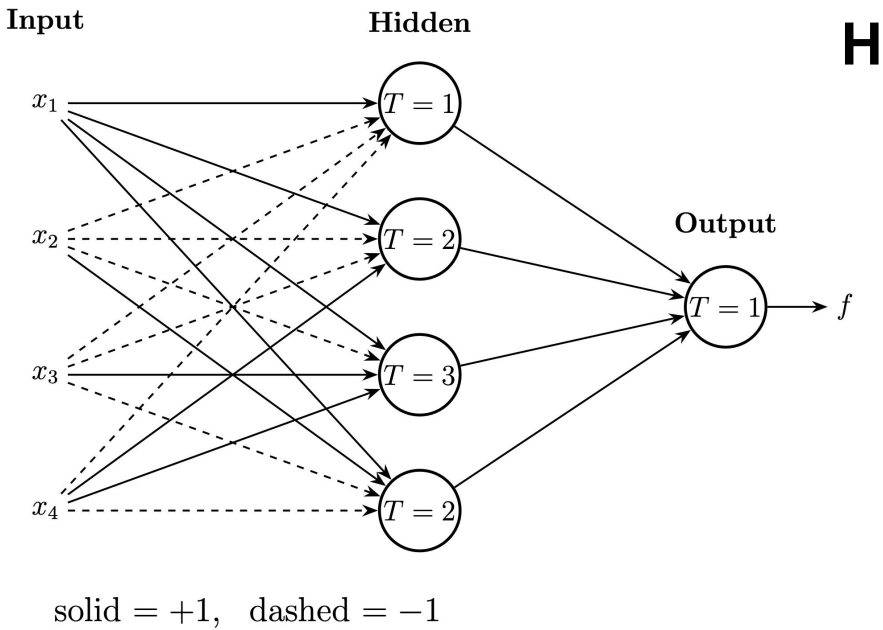
$$f = (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)$$

solid $= +1$, dashed $= -1$

One-hidden-layer MLP is a Universal Boolean Function

Universal Boolean Function

One-hidden-layer MLP for a boolean function with N inputs:



How many neurons do we need?

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

Universal Boolean Function

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$
\begin{aligned}
f = & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)
\end{aligned}
$$

Karnaugh Map (K-map)

x_1x_2 \ x_3x_4	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	**1**	0	0	0
10	**1**	**1**	**1**	0

Universal Boolean Function

x_1	x_2	x_3	x_4	$f(x_1, x_2, x_3, x_4)$
1	0	0	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	1

$$
\begin{aligned}
f = & (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, \bar{x}_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, \bar{x}_2 \,\&\, x_3 \,\&\, x_4) \\
& \mid (x_1 \,\&\, x_2 \,\&\, \bar{x}_3 \,\&\, \bar{x}_4)
\end{aligned}
$$

Karnaugh Map (K-map)

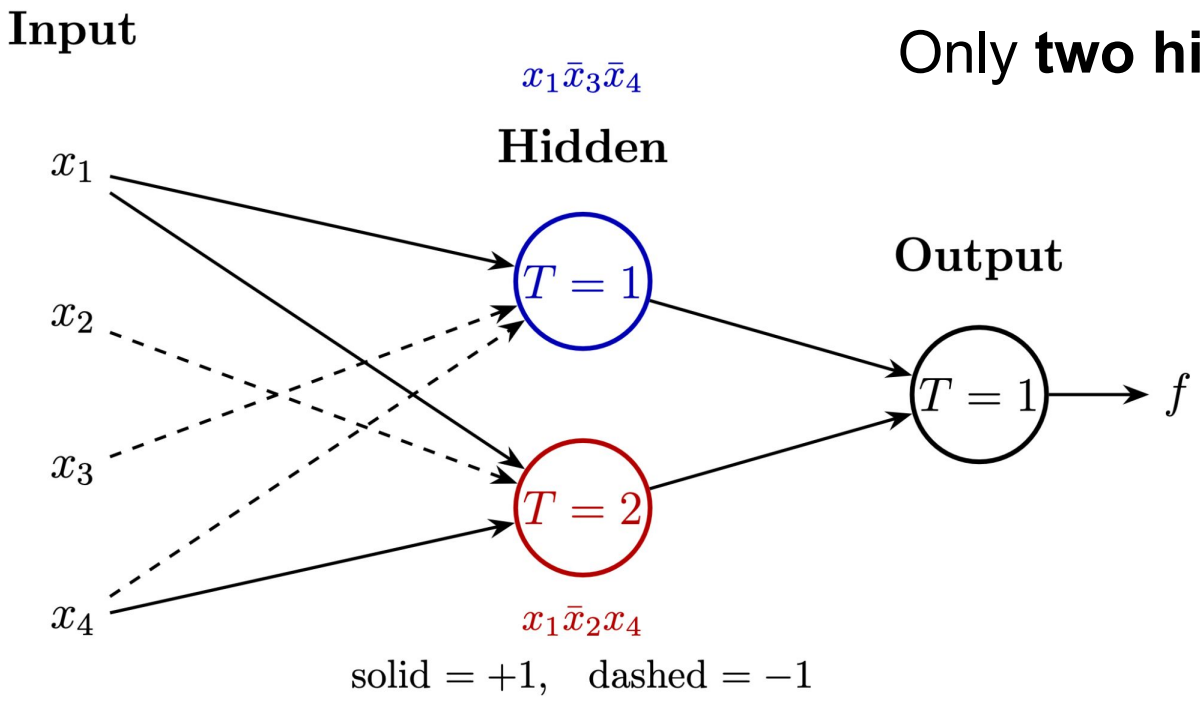
x_1x_2 \ x_3x_4	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	0	0	0
10	1	1	1	0

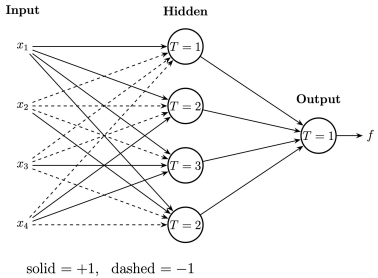
Reduced DNF $f = x_1\bar{x}_3\bar{x}_4 \mid x_1\bar{x}_2x_4$

Universal Boolean Function

$$f(x_1, x_2, x_3, x_4) = \Big((x_1 \,\&\, \bar{x}_2 \,\&\, x_4) \mid (x_1 \,\&\, \bar{x}_3) \Big) \,\&\, \Big((x_2 \,\&\, x_3) \mid \overline{x_2 \,\&\, x_4} \Big)$$

Only **two hidden neurons** needed





Universal Boolean Function

One-hidden-layer MLP for a boolean function with N inputs:

Worst-case K-map

x_1x_2 \ x_3x_4	00	01	11	10
00	0	**1**	0	**1**
01	**1**	0	**1**	0
11	0	**1**	0	**1**
10	**1**	0	**1**	0

No adjacent 1s

Need **up to 2^{N-1}** hidden neurons

Exponential in N

$$f = x_1 \oplus x_2 \oplus x_3 \oplus x_4$$

Summary

- One-hidden-layer MLP is a Universal Boolean Function
- However, it could require an exponential number of neurons